

CSA4001-MANAGEMENT INFORMATION SYSTEM FOR DATA OPTIMIZATION

NAME: k.karthik

REG.NO.: 192110178

4. Draw a UML diagram for ATM System using CASE tool. The banking system allows a customer to access the financial transactions by ATM System, it has a step-by-step process describe the work of this process and elaborate the what are the work can do by customer, banking system, administrator and technicians with the ATM system.

AIM

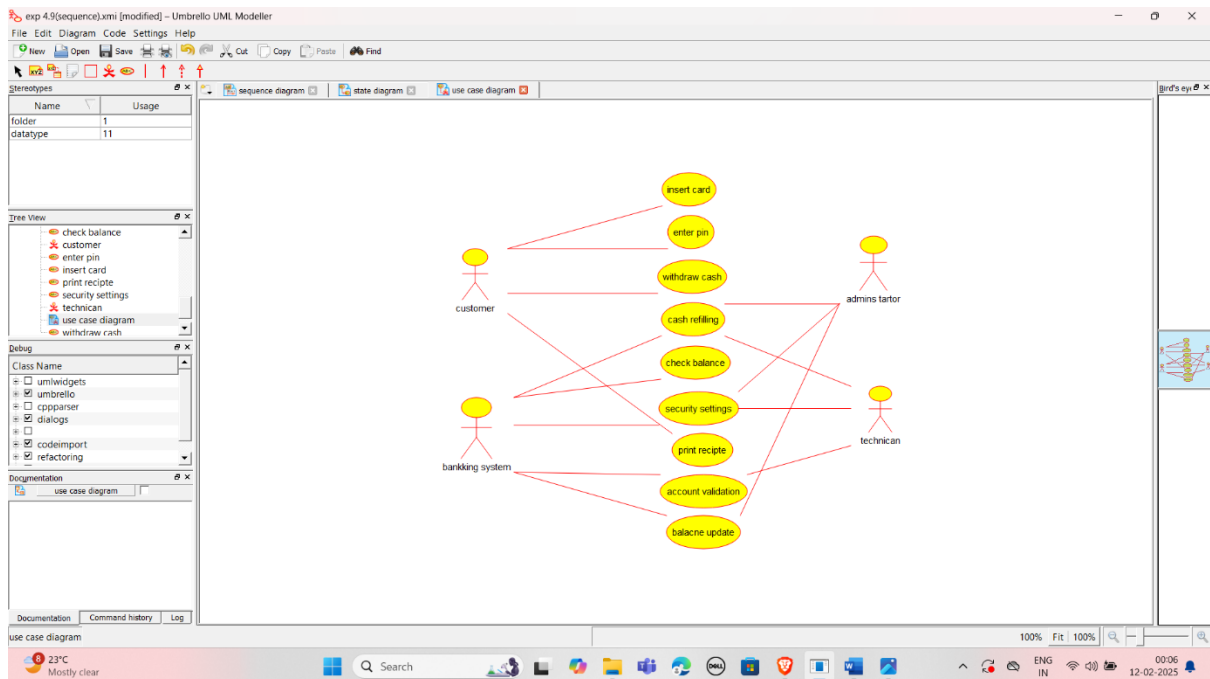
To develop a Use Case Diagram for an ATM System, modelling the interactions between Customers, Banking System, Administrators, and Technicians to perform financial transactions and system maintenance.

PROCEDURE:

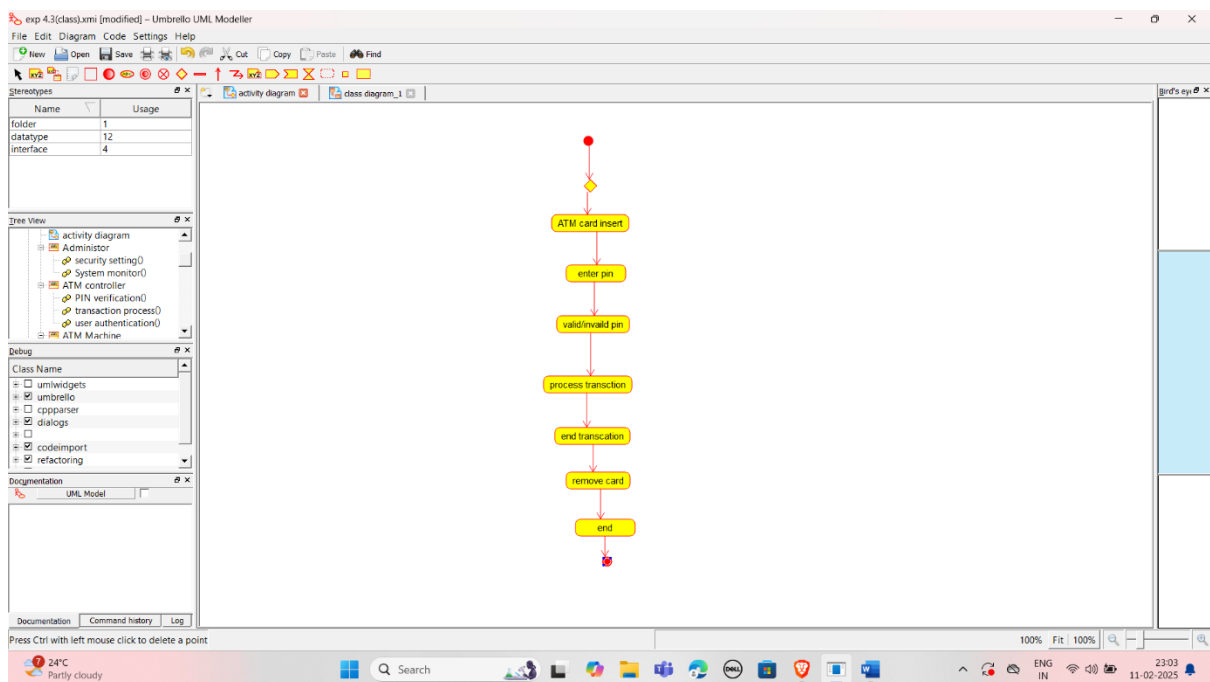
1. Identify the actors: Customer (performs transactions), Banking System (processes transactions), Administrator (manages ATM security and configuration), and Technician (maintains the ATM system).
2. Identify key use cases: Insert Card, Enter PIN, Withdraw Cash, Deposit Cash, Transfer Funds, Check Balance, Print Receipt, Validate PIN, Process Transaction, Update Account, Generate Transaction Report, Monitor ATM, Configure ATM, Enable/Disable ATM, Perform Maintenance, Refill Cash, Fix ATM Issues.
3. Define relationships: Withdraw Cash, Deposit Cash, Transfer Funds, and Check Balance include Enter PIN for authentication; Process Transaction is linked to Banking System; Administrator manages ATM configurations; Technician handles ATM maintenance.
4. Draw the system boundary labelled ATM System, placing actors outside and use cases inside.
5. Connect actors to their respective use cases using association lines to represent interactions.
6. Use include and extend relationships where necessary, such as Enter PIN being included in all transaction-related use cases.
7. Verify the completeness and correctness of the diagram to ensure it accurately represents ATM functionalities and interactions.

OUTPUT:

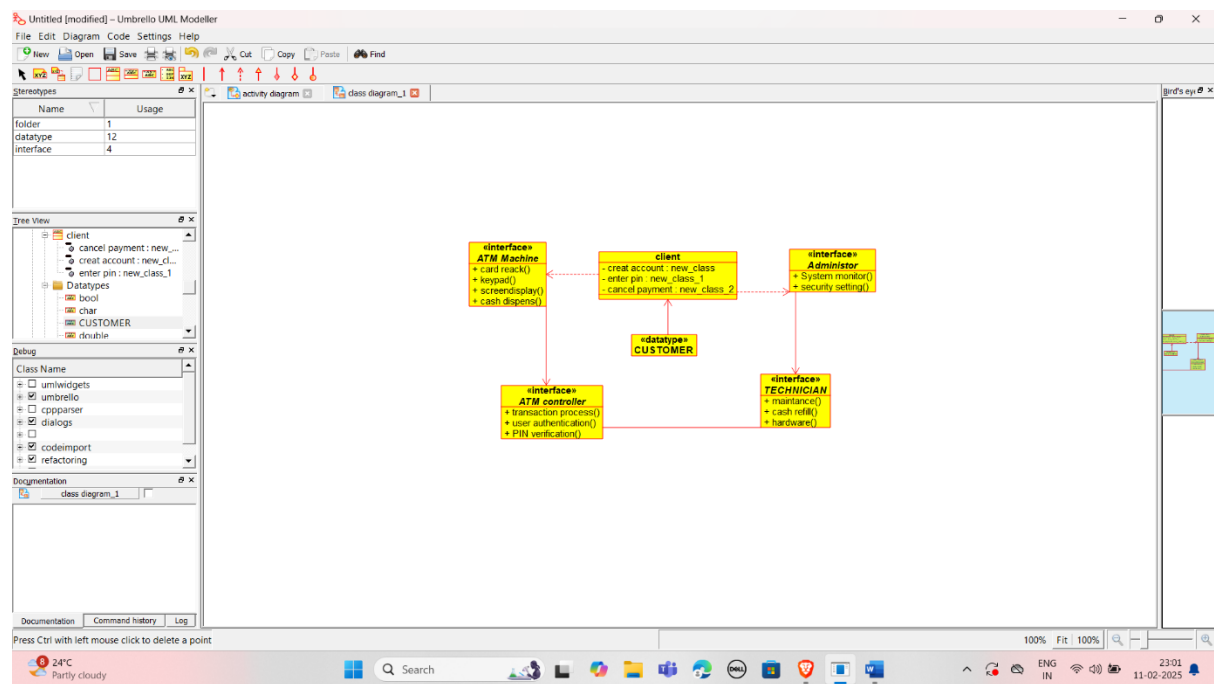
USE CASE DIAGRAM:



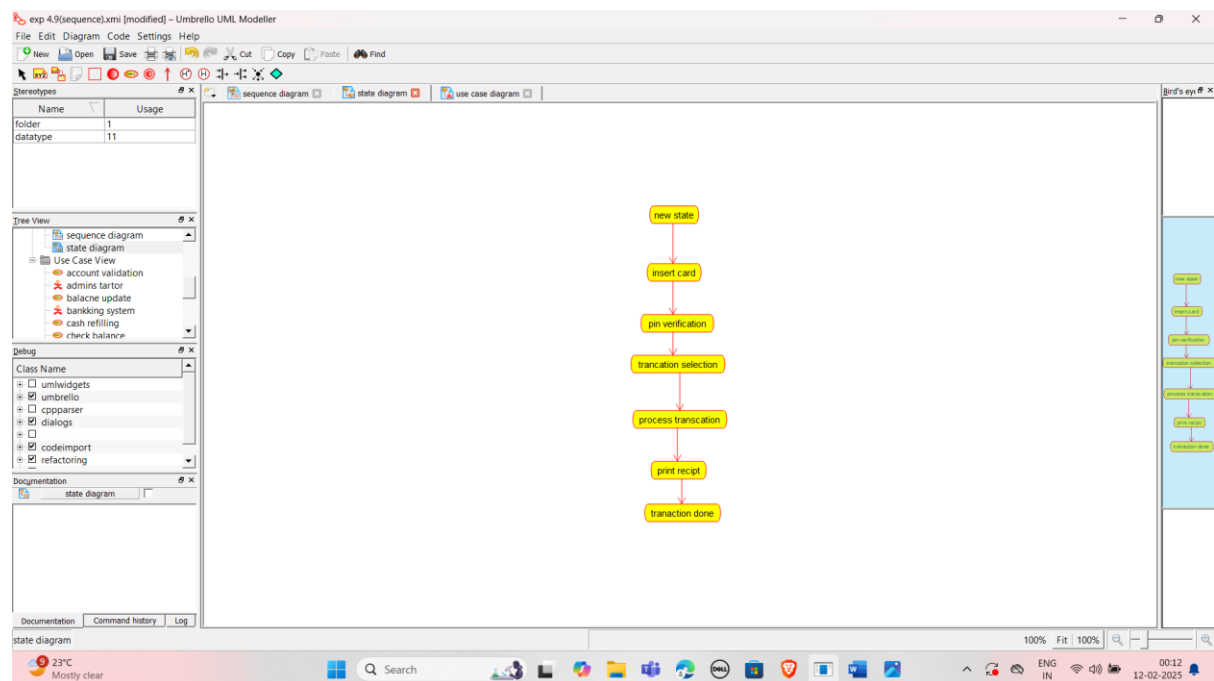
ACTIVITY DIAGRAM:



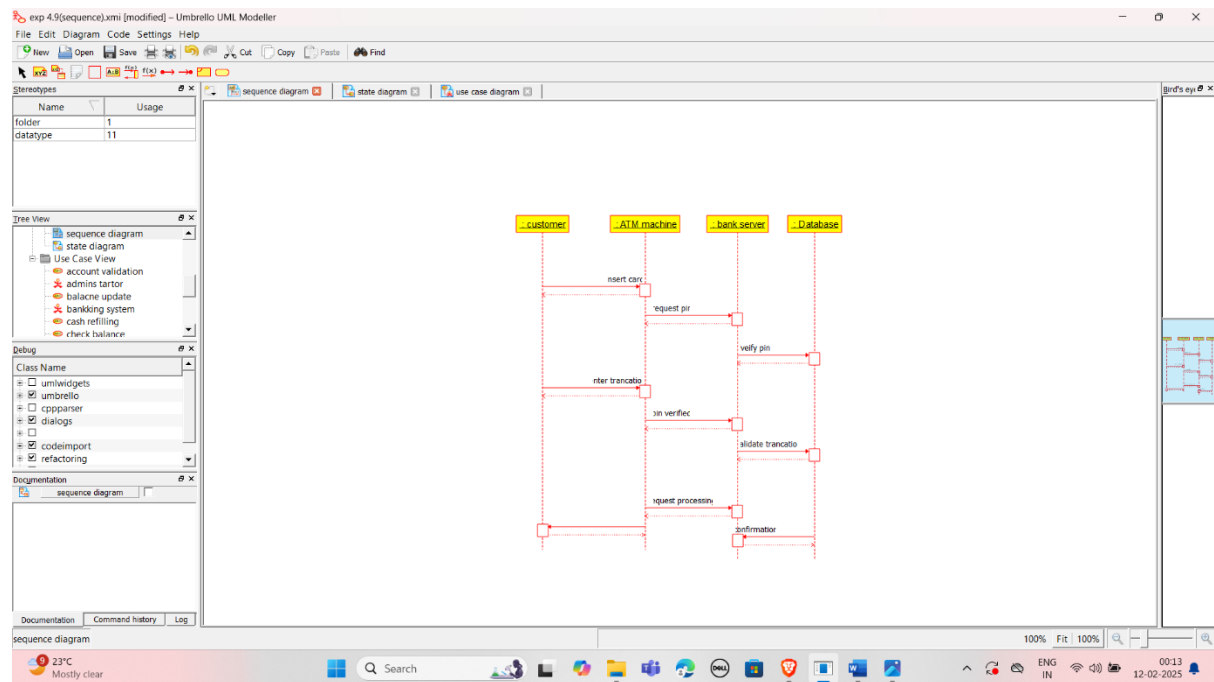
CLASS DIAGRAM:



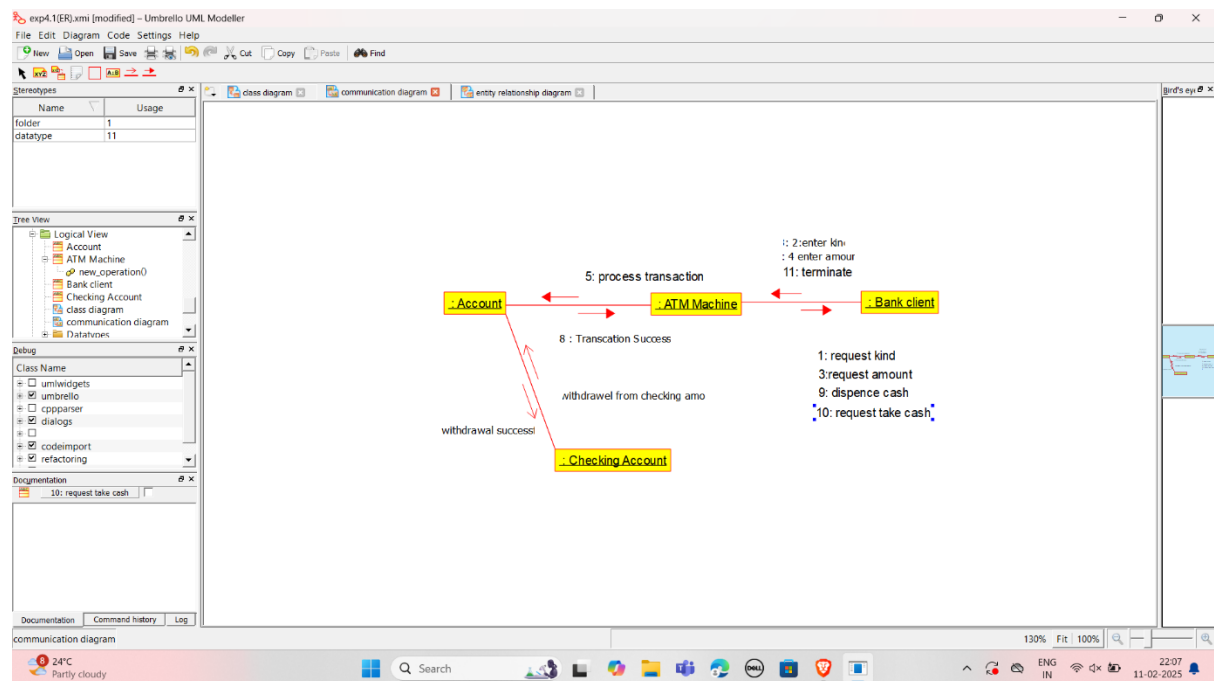
STATE DIAGRAM:



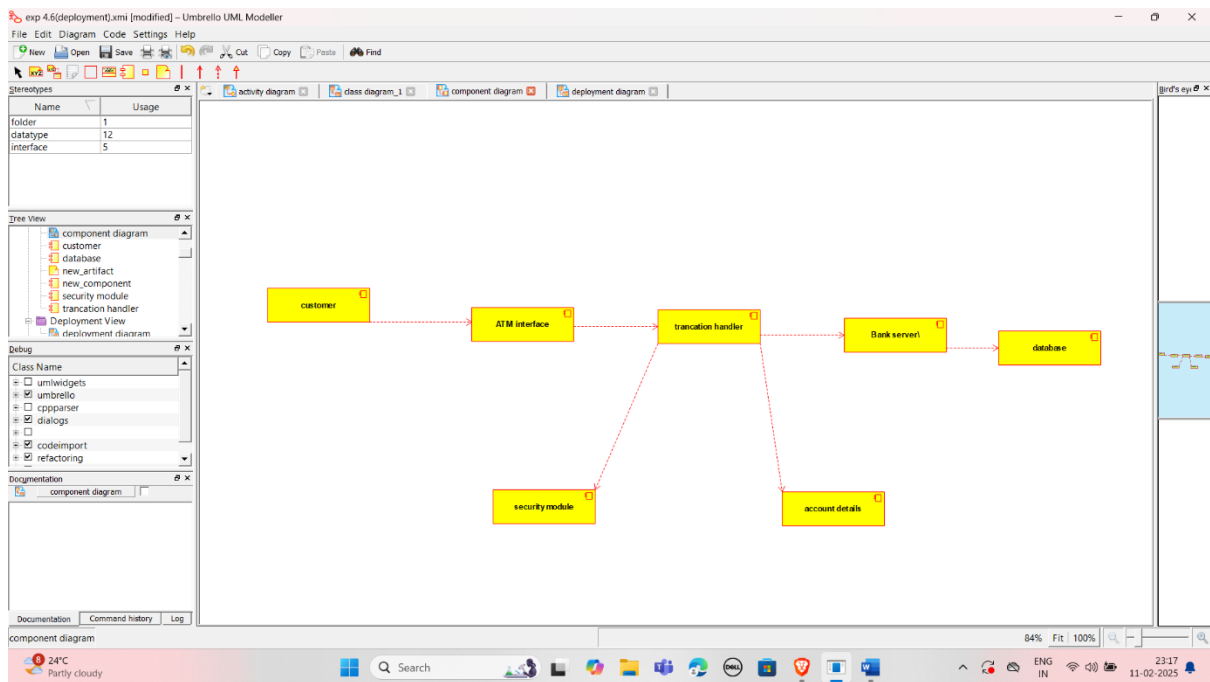
SEQUENCE DIAGRAM:



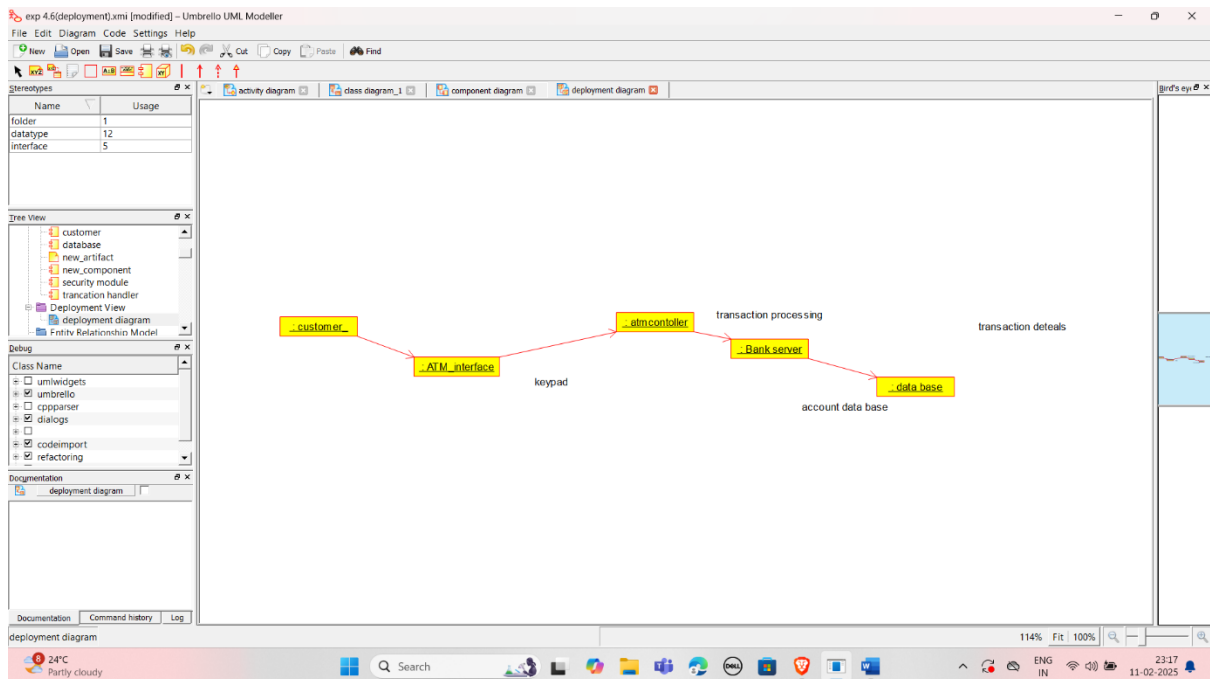
COMMUNICATION DIAGRAM:



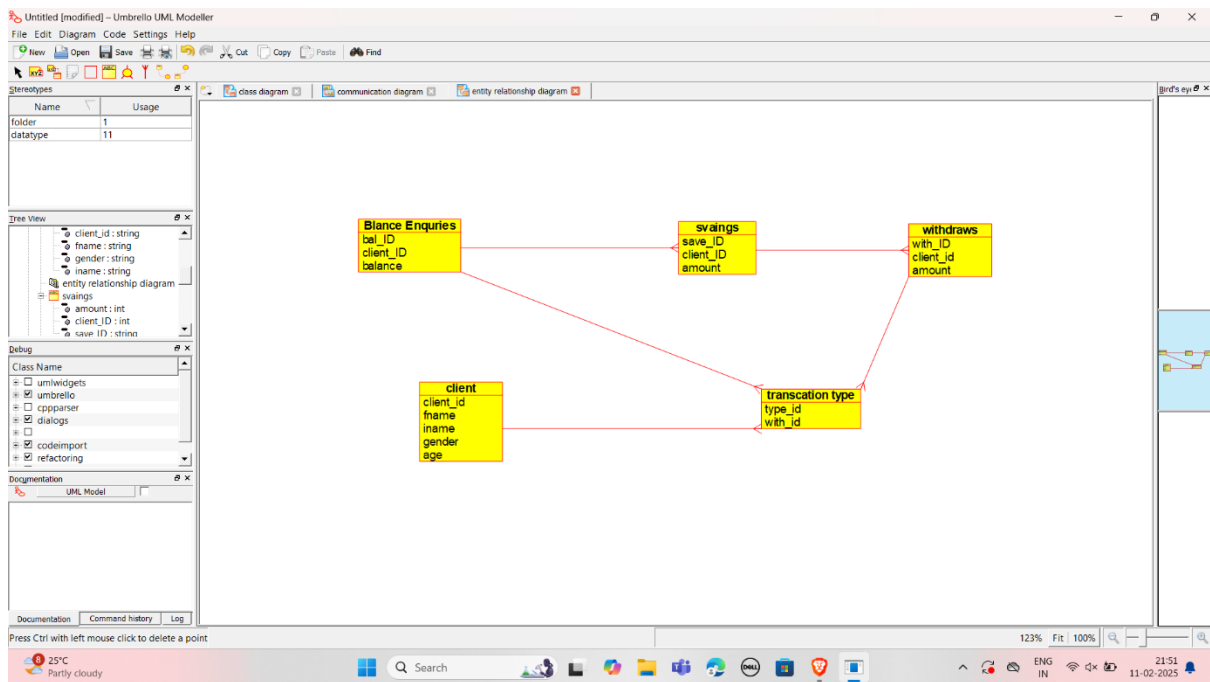
COMPONENT DIAGRAM:



DEPLOYMENT DIAGRAM:



ENTITY RELATIONSHIP DIAGRAM:



RESULT:

The Use Case Diagram for the ATM System clearly models user interactions, transactions, and maintenance, serving as a structured blueprint for system design.